Manual

Multiple Assignment of Resources

HLS-MFB 8.1

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1 Overview of Multiple Assignment of Resources

Purpose

Capacity inspection and automatic assignment assumes in HYDRA shop floor terminal that only one order/ operation can run at any one time. If a multiple assignment made manually, a dialog appears making the user aware of the double assignment; automatic assignment always operates on the premise of a single assignment.

Different workplaces or rather machines provide the ability for operations to be multiply assigned (parallel assignment). Examples for this are pallet machines, furnaces, assembly workplaces.

Because of the nature of the technology, however, the scope of available multiple assignment options can certainly vary. For example, a pallet machine has the capability of processing five orders at the same time, while where a furnace is concerned, the assignment may depend on how large the articles are and at what temperature the articles can be burned. At assembly workplaces, the use and as such the assignment may depend on the dimensions of the parts/ products being assembled.

You make use of this functions package, if these kinds of application scenarios are found in your production and they need to be accounted for accordingly in the HYDRA shop floor terminal planning process.

Integration

The results of planning are shown in the order sequencing list at the shop floor terminal.

Features

- Function used to support a (parallel) multiple assignment of workplaces/ machines (e.g. furnaces, assembly workplaces) in HYDRA shop floor terminal's graphic planning board
- Definition of the workplace/ machine-related availability (available capacities)
- Calculation of requirements based on factors defined at the operation, such as required space

2 Configuration Multiple Assignment of Resources

Overview

The planning component within HYDRA shop floor terminal is currently based on the assumption of no shift-free times with a fixed availability of 1000 at any one workplace. An operation's requirement is currently also set fix at 1000. So, when an operation is assigned, the availability of the workplace is used up entirely (availability 1000 - requirement 1000 = availability 0).

To allow for greater flexibility within the HYDRA shop floor terminal, the availability can now be changed on one side and the requirement can be changed on the other side.

Described below are the steps with which to activate multiple assignments in HYDRA shop floor terminal.

Workplace availability

At the workplace (workplace/ machine configuration, index tab HLS), availability is defined in one field.

The valid range of values for the availability field is between 1000-99999.

Configuration for defining the requirement

Configuring the order type

The requirement is defined via the operation. To assure the highest possible flexibility, a formula can be defined at the operation that is then used to calculate the requirement for an operation. The reference to a formula for this is defined to one of the operation's user fields so that the customer can transfer it via the interface. Which particular user field contains the reference to the formula is defined by specific order type (order type configuration). The formula itself is defined in HYDRA in formula management.

The formula defined in the operation's specified user field calculates its requirement. Possible values for the user field are 45-66.

Creating a formula

The formula is defined in HYDRA formula management that is meant to calculate the requirement for the operation.

Currently, the following user fields can be used in a formula to calculate the requirement: ANR.FU:23, ANR.FU:24, ANR.FU:25, ANR.FU:26, ANR.FU:27, ANR.FU:28.

Configuring user fields

Depending on the formula, the user fields used must be added to the operation.

In this example, the formula is stored in user field 45 (see configuring order types). Now user field 45 must be configured for the operation.

User fields 23 and 24 are used in the formula "REQUIREMENT", i.e. these must also be configured at the OP.

Defining the requirement at the operation

The values defined at the operation are used to determine the requirements.

If no formula is defined at the operation, a requirement of 1000 is assumed. This is the simplest of scenarios, because with it, multiple assignments can be defined based on the number of operations. For this purpose, a multiple of 1000 is defined as availability. The operation displayed above would have an requirement of 2000.